University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-4 Odd Semester Examination-2024
Course: CSE4141 (Microprocessor Interfacing and Microcontrollers)
Marks: 52.50 Time: 3:00 Hours

[N.B. Answer any Six (06) questions taking Three (03) from each section]

Section-A

1. (a) With a suitable example illustrate steps for designing and integration of peripheral devices to a computer. 2.00
   (b) What do you mean by address decoding? What are the needs of address decoding? 2.00
   (c) Discuss peripheral-mapped I/O and memory-mapped I/O with necessary control signals and instruction. 3.00
   (d) Distinguish between absolute and partial address decoding. 1.75

2. (a) Draw the timing-waveform of the execution of instruction OUT 01H. 2.75
   (b) How many memory locations can be addressed by a microprocessor with 14 address lines? How many address lines are necessary to address two megabytes (2048K) of memory? 2.00
   (c) Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch as an output port. Also write the program to display digit 5 to the output port. 4.00

3. (a) Discuss an interfacing circuit to design a safety control system for home appliances using a decoder, a buffer as an input port and a latch as an output port. Also write the program to operate the controlling system. 4.00
   (b) Draw an interfacing circuit to monitor various processes (conveyer belts) on the floor of a manufacturing plant and write a program to perform the following operations: 4.75
   (i) Turn on the five conveyer belts according to the ON/OFF positions of switches $S_4 - S_0$ at port F1H.
   (ii) Turn off the conveyer belts and turn on the emergency alarm only when both switches – S_7 from port F1H and S_7'from port F2H – are triggered.
   (iii) Monitor the switches continuously.

4. (a) What do you mean by Programmable Peripheral Interfacing (PPI)? 1.00
   (b) Shortly discuss the I/O Ports and Timer of 8155 with block diagram. 3.75
   (c) Design a square-wave generator with a pulse width of 100μs by using the 8155 timer. Set up the timer in Mode 1 if the clock frequency is 3 MH_z. 4.00

Section-B

5. (a) Discuss 8255A control word formats. 3.00
   (b) Discuss BSR mode of 8255A with a suitable example. 3.00
   (c) Interface 8255A ports in mode 0 with processor for the following conditions: 2.75
   (i) Use port B and port C_L as input ports to read inputs from DIP switches and port A and port C_U as output ports to display output at leds.
   (ii) Write the program to read the DIP switches and display the reading from port B at port A and from port C_L at port C_U.

6. (a) Draw the ARM register set organization and explain the role of each category of registers. 6.00
   (b) Differentiate between register R13 (SP) and register R14 (LR) in ARM architecture in terms of their working principle. 2.75

7. (a) Suppose you want to send the message "CSE" from one microcontroller to another microcontroller. You must confirm receipt of the data byte to the sender. The sender address is (130)10 and the receiver address is (130)10. Which communication protocol would be most suitable for this purpose. Design the packet format for that protocol and describe the protocol step by step with proper justification and illustration. 5.75

   (b) If the baud rate is 9600 bps, how many characters can be transmitted in one second (assuming 8 data bits, 1 start, 1 stop)? Choose proper communication protocol for this purpose. 3.00

8. (a) Suppose you want to connect three microcontrollers for data exchange. Which protocol (UART, SPI, or I²C) would you choose and why? Describe the protocol with proper justification. 4.00

   (b) Suppose, two microcontrollers use the SPI communication protocol. If the clock frequency is 4 MHz, how long will it take to transmit 32 bits of data? 2.75

   (c) Explain why pull-up resistors are needed in I²C lines. 2.00

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[N.B. Answer any Six questions taking Three from each section]

Section-A

1.	(a)	Explain PUSH and POP instruction of Cortex M3 or M4 with examples.	3.5
	(b)	Explain LDR and STR Instructions of Cortex M3 or M4 with examples	3.25
	(c)	Explain MOVT Instruction of Cortex M3 or M4 with examples.	2
2.	(a)	Load the roll numbers of team leaders of five teams of your class into register R0 to R4. Write a register-level program for a STM32F103X or STM32F446X, (i) to discard the session and hall code from the roll number (ii) move the roll numbers stored in R0~R4 to a SRAM area, starting address is 0x2FF20F00.	3.25
	(b)	Assume there are 100 integer numbers stored in SRAM area, starting address is 0x2FF20000, with each integer having a value ranging from 0 to 255. Write a at register-level program for STM32F103X or STM32F446X to sort these numbers in ascending order using Selection sort.	5.5
3.	(a)	Load the roll numbers of team leaders of five teams of your class into register R0 to R4. Write a register-level program for STM32F103X or STM32F446X, (i) to move the roll numbers stored in R0~R4 to SRAM area, starting address is 0x2FF20400, (ii) then move all roll numbers to SRAM area, starting address is 0x2FF20500	3.5
	(b)	Store the ASCII codes of uppercase-letters into SRAM area staring from 0x2FF10000. Write a register-level program for STM32F103X or STM32F446X to convert uppercase-letters into corresponding lowercase-letters and store their ASCII codes into SRAM area, starting from 0x2FF20000.	5.25
4.		You have given four sets of positive integer number that are stored in the SRAM area, the starting addresses and number of integers for each set are specified as follows: Set 1: Starting at address 0x2FF20000, contains 18 integers. Set 2: Starting at address 0x2FF20600, contains 09 integers. Set 3: Starting at address 0x2FF20900, contains 21 integers. Set 4: Starting at address 0x2FF20A00, contains 06 integers. Write a program for STM32 Cortex processor to determine the largest and smallest values from each set and store these values sequentially in memory.	8.75

Section-B

5.	(a)	Draw the block diagram of ARM Cortex M4 and describe its major component.	5
	(b)	What are the main features of Floating Pont Unit (FPU) of Cortex M4.	3.75
6.	(a)	Explain the role of SDA and SCL lines in I2C communication protocol.	2
	(b)	What are the advantages and disadvantages of using I2C over SPI protocol?	2
	(c)	Describes the different fields of Message of I2C protocol.	4.75
7.	(a)	Describe the functions of MOSI, MISO, SCLK, and SS on SPI protocol.	3
	(b)	Illustrate with block diagram, how multiple Slaves could communicate with Master, when master has a single SS/CS line and has multiple separate SS/CS lines.	3
	(c)	Explain how full-duplex communication can be performed in SPI protocols.	2.75
8.	(a)	Describe the structure of a UART data frame. Describe details of start bit, data bits, parity bit, and stop bits.	3
	(b)	Explain the steps of how flow control is managed in UART communication protocol.	3.75
	(c)	What are the roles of 12th bit of USART_CR1; 6th and 7th bits of USART_SR of STM32F446?	2

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-4 Odd Semester Examination-2023
Session: 2018-2019
Course: CSE4141 (Computer Peripherals and Interfacing)
Marks: 52.50 Time: 3:00 Hours

[N.B. Answer any Six questions taking Three from each section]

Section-A

1.	(a)	What is meant by computer peripheral interfacing? Explain with example.	3
	(b)	Distinguish between port-addressed I/O and memory-mapped I/O.	3.75
	(c)	How many I/O devices can be interfaced with port-addressed I/O? Explain.	2
2.	(a)	Draw the timing-waveform of the execution of instruction OUT 01H.	2.75
	(b)	Discuss absolute address decoding and partial address decoding with example.	3
	(c)	Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch as an output port. Also write the program to display digit 7 to the output port.	3
3.	(a)	Discuss how time delay is calculated using register pair technique.	4
	(b)	Discuss an interfacing circuit to design a safety control system for home appliances using a decoder, a buffer as an input port and a latch as an output port. Also write the program to operate the controlling system.	4.75
4.	(a)	Discuss how to interface eight DIP Switches using 74LS138 Decoder and 74LS244 Tri-State Octal Buffer.	3.75
	(b)	Discuss how does a microprocessor differentiate between data (operand) and instruction code (opcode)	2
	(c)	Discuss with example how CALL and RET instructions work in between a main program and a subroutine.	3

Section-B

5.	(a)	What do you mean by maskable and nonmaskable interrupt of 8085?	2
	(b)	Can the microprocessor be interrupted again before the completion of the first interrupt service routine? Explain.	2
	(c)	Discuss the 8085 vectored interrupts with necessary diagram.	4.75
6.	(a)	Discuss how an 8-bit A/D converter is interfaced with processor using status check I/O technique.	3.75
	(b)	Design an output port with the address FFH to interface the 1408 D/A converter that is calibrated for 0 to 10V range. Write a program to generate a continuous ramp waveform. Explain the operation of the 1408 for the calibration of a bipolar range +5V to -5V. Calculate the output V_o if the input is 10000000_2.	5
7.	(a)	What do you mean by Programmable Peripheral Interfacing (PPI)? Explain.	1.75
	(b)	Discuss the control word format of 8155.	3
	(c)	Design a square-wave generator with a pulse width of 100μs by using the 8155 timer. Set up the timer in Mode 1 if the clock frequency is 3 MH_z.	4
8.	(a)	Discuss 8255A control word formats.	3
	(b)	Discuss BSR mode of 8255A with a suitable example.	3
	(c)	Interface 8255A ports in mode 0 with processor for the following conditions: (i) Use port B and port C_L as input ports to read inputs from DIP switches and port A and port C_U as output ports to display output at leds. (ii) Write the program to read the DIP switches and display the reading from port B at port A and from port C_L at port C_U.	2.75

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Full Marks: 52.5 **Duration: 3 Hours**

[N.B. Answer any SIX questions taking THREE from each of the section]

Section- A

1. a) What is meant by computer peripheral interfacing? Explain with example. 2.75
   b) Distinguish between port-addressed I/O and memory-mapped I/O. 3
   c) Draw the timing-waveform of the execution of instruction IN 84H. 3

2. a) Is it possible in interfacing, IN 20H and then OUT 20H? Explain why or why not. 2
   b) Explain absolute address decoding and partial address decoding with example. 3
   c) Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch as an output port. Also write the program to display digit 5 to the output port. 3.75

3. a) Explain why handshake signals are used to interface I/O devices with processor. 2
   b) Discuss how maximum time delay is calculated in 8085 using a register pair with a nested loop. 3
   c) Discuss an interfacing circuit to design a safety control system for home appliances using a decoder, a buffer as an input port and a latch as an output port. Also write the program to operate the controlling system. 3.75

4. a) How long can the INTR pulse stay high? 2
   b) Discuss the 8085 vectored interrupts with necessary diagram. 4.75
   c) Explain the use of SIM and RIM instructions. 2

Section- B

5. a) Discuss how an 8-bit A/D converter is interfaced with processor using status check I/O technique. 4
   b) Design an output port with the address FFH to interface the 1408 D/A converter that is calibrated for 0 to 10V range. Write a program to generate a continuous ramp waveform. Explain the operation of the 1408 for the calibration of a bipolar range +5V to -5V. Calculate the output Vo if the input is 10000000_2. 4.75

6. a) What is meant by PPI? 1
   b) Discuss the control word format of 8155. 4
   c) Design a square-wave generator with a pulse width of 100μs by using the 8155 timer. Set up the timer in Mode 1 if the clock frequency is 3 MHz. 3.75

7. a) Why BSR mode of 8255A is used? Give an example. 2
   b) Interface 8255A ports in mode 0 with processor for the following conditions: 3
      i. Use port B and port C_L as input ports to read inputs from DIP switches and port A and port C_U as output ports to display output at leds.
      ii. Write the program to read the DIP switches and display the reading from port B at port A and from port C_L at port C_U.
   c) Draw and discuss the block diagram of 8254 programmable interval timer. 3.75

8. a) Discuss how data is transferred bidirectionally between two computers using port A of 8255A in Mode 2. 3
   b) What is key debounce? How many ways it can be generated? Explain. 2
   c) How a 20-key matrix keyboard is interfaced with processor using an 8255A PPI. 3.75

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[N.B. Answer any **Six** questions taking **Three** from each section]

Section-A

1. a) What do you mean by computer peripheral interfacing? 2
   b) Discuss port-addressed I/O and memory-mapped I/O with control signals and instructions. 4.75
   c) Can it be- LDA8000H and then STA8000H? Explain why or why not. 2

2. a) Define absolute address decoding and partial address decoding with example. 3
   b) Draw the timing-waveform to execute the instruction OUT 01H. 2
   c) Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch as an output port. Also write the program to display digit 8 to the output port. 3.75

3. a) Discuss how time delay is calculated using register pair technique with a counter value of FFFFH. 4
   b) Discuss an interfacing circuit to design a safety control system for home appliances using a decoder, a buffer as an input port and a latch as an output port. Also write the program to operate the controlling system. 4.75

4. a) Differentiate maskable and non-maskable interrupts. 2
   b) Can the microprocessor be interrupted again before the completion of the first interrupt service routine? Why or why not? 2
   c) Is there a minimum pulse width required for the INTR signal? Explain. 1.75
   d) Write a main program to count continuously in binary with a one-second delay between each count and write a service routine to flash FFH six times when the program is interrupted, with some appropriate delay between each flash. 3

Section-B

5. a) What is vectored interrupt? Discuss the vectored interrupts of 8085 processor. 5
   b) Design a 1-minute timer using a 60Hz power line as an interrupting source. The output ports display minutes and seconds in BCD. At the end of the minute, the output ports should continue displaying one minute and zero seconds. 3.75

6. a) Discuss how an 8-bit A/D converter is interfaced with processor using status check I/O technique. 3.75
   b) Design an output port with the address FFH to interface the 1408 D/A converter that is calibrated for 0 to 10V range. Write a program to generate a continuous ramp waveform. Explain the operation of the 1408 for the calibration of a bipolar range +5V to -5V. Calculate the output V_o if the input is 10000000_2. 5

7. a) What is PPI? Write down the 8255A PPI control word format for BSR mode. 2
   b) What is Control Word? Determine the control word for the following configuration of 8255:- 2.75
   Port A –Output
   Mode of port A –Mode 1
   Port B –Output
   Mode of port B –Mode 0
   Port C lower (pins PC0 –PC2) –Output
   c) Use 8255A in mode 1 where a printer is interfaced using port B with status I/O. Write initialization instructions and a printer subroutine to output characters those are stored in memory. 4

8. a) Discuss BSR mode of 8255A with a suitable example. 3
   b) Interface 8255A ports in mode 0 with processor for the following conditions: 3
      (i) Use port B and port C_L as input ports to read inputs from DIP switches and port A and port C_U as output ports to display output at leds.
      (ii) Write the program to read the DIP switches and display the reading from port B at port A and from port C_L at port C_U.
   c) Briefly discuss how a counter of 8254 is initialized and the value of the counter is read during counting. 2.75

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Marks: 52.5 Time: 3:00 Hours

[N.B. Answer any ***three*** questions from each section]

Section - A

1.	(a)	Define computer peripheral interfacing with example.	3
	(b)	Compare port-addressed I/O and memory-mapped I/O on basis of address length, control signal, and instruction.	3.75
	(c)	In interfacing program, is it possible to write, IN 20H and then OUT 20H? Explain why or why not.	2
2.	(a)	Distinguish between absolute address decoding and partial address decoding.	3.75
	(b)	Discuss how device select pulse is generated prior to transfer data in between processor and I/O devices.	3
	(c)	Draw the circuitry to interface eight DIP switches using a 74LS138 decoder and a 74LS244 buffer.	2
3.	(a)	Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch. Also write the program to display digit 7 to the output port.	4.75
	(b)	Draw an interfacing circuit to design a safety control system for some home appliances using a decoder, a buffer and a latch. Also write the program to operate the controlling system.	4
4.	(a)	Describe the delay calculation process using register pair.	3
	(b)	Discuss 8085 vectored interrupts.	5.75

Section - B

5.	(a)	What do you mean by maskable and nonmaskable interrupts?	2
	(b)	Write a main program to count continuously in binary with a one-second delay between each count and write a service routine at XX70 to flash FFH five times when the program is interrupted, with some appropriate delay between each flash.	3
	(c)	Write the difference between vectored and non-vectored interrupts.	1.75
	(d)	Can a processor be interrupted again before the completion of the first interrupt service routine? Explain why or why not.	2
6.	(a)	Discuss how an 8-bit A/D converter is interfaced with processor using status check I/O technique.	4
	(b)	Design an output port with address FFH to interface the 1408 D/A converter that is calibrated for 0 to 10V range. Write a program to generate a continuous ramp waveform.	4.75
7.	(a)	Describe the functional diagram of 8257 DMA controller.	4.75
	(b)	Draw and discuss the block diagram of 8254 programmable interval timer.	4
8.	(a)	What is meant by Programmable Peripheral Interfacing (PPI)?	1
	(b)	Design a square-wave generator with a pulse width of 100μs by using the 8155 timer. Set up the timer in Mode 1 if the clock frequency is 3 MH_z.	3.75
	(c)	Interface 8255A ports in mode 0 with processor for the following conditions: (i) Use port B and port C_L as input ports to read inputs from DIP switches and port A and port C_U as output ports to display output at LEDs. (ii) Write the program to read the DIP switches and display the reading from port B at port A and from port C_L at port C_U.	3
	(d)	Why the BSR mode of 8255A is used?	1

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[N.B. Answer any Six questions taking Three from each section]

Section-A

1.	(a)	What is meant by computer peripheral interfacing? Explain with example.	3
	(b)	Distinguish between port-addressed I/O and memory-mapped I/O.	3.75
	(c)	Is it possible in interfacing, IN 40H and then OUT 40H? Explain why or why not.	2
2.	(a)	Handshake signals are used to interface I/O devices with processors. Can you explain why and how these signals are used?	4
	(b)	What do you know about the term 'MPU'? Is 8085 an MPU? Why or why not?	4.75
3.	(a)	Discuss how maximum time delay is calculated in 8085.	4
	(b)	Discuss an interfacing circuit to design a safety control system for home appliances using a decoder, a buffer as an input port and a latch as an output port. Also write the program to operate the controlling system.	4.75
4.	(a)	Define maskable, nonmaskable, vectored, and non-vectored interrupts of 8085.	4
	(b)	Is there a minimum pulse width required for the INTR signal?	1.75
	(c)	Write a main program to count continuously in binary with a one-second delay between each count and write a service routine at XX70 to flash FFH five times when the program is interrupted, with some appropriate delay between each flash.	3

Section-B

5.	(a)	Why SIM and RIM instructions are used?	2
	(b)	What is NMI? Mention the difference between RST 7.5 and RST 7.	2.75
	(c)	Design a 1-minute timer using a 60Hz power line as an interrupting source. The output ports display minutes and seconds in BCD. At the end of the minute, the output ports should continue displaying one minute and zero seconds.	4
6.	(a)	What do you know about the I/O Ports and Timer of 8155.	4.75
	(b)	Is it possible to design a square-wave generator with a pulse width of 100μs by using the 8155 timers? If your answer is 'yes' then explain how? If your answer is 'No' then explain why.	4
7.	(a)	What is meant by PPI?	1
	(b)	Discuss the control word format of 8155.	4
	(c)	Discuss how an 8-bit A/D converter is interfaced with processor using status check I/O technique.	3.75
8.	(a)	Why BSR mode of 8255A is used? Give an example.	2
	(b)	Interface 8255A ports in mode 0 with processor for the following conditions: (i) Use port B and port C_L as input ports to read inputs from DIP switches and port A and port C_U as output ports to display output at LEDs. (ii) Write the program to read the DIP switches and display the reading from port B at port A and from port C_L at port C_U.	3
	(c)	Draw and discuss the block diagram of 8254 programmable interval timer.	3.75